Secure a New Ubuntu Server

Audit, Configuration, Hardening and Monitoring of secure-server (Docker + systemd)

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Lab environment	Windows host, Docker Desktop (WSL 2), Ubuntu 24.04 container with systemd as PID 1
Server hostname	secure-server
Final SSH access	ssh -i ~/.ssh/id_ed25519 admin@localhost -p 2222

1. Lab Environment and Users

An Ubuntu 24.04 container named secure-server was built from a Dockerfile and run with systemd as PID 1, so that systemctl and journalctl behave like a real server. SSH was exposed to the host on port 2222.

1.1 Build and run

```
docker build -t secure-server-img .
docker run -d --name secure-server --hostname secure-server --privileged \
  --cgroupns=host -v /sys/fs/cgroup:/sys/fs/cgroup:rw -p 2222:22 secure-server-img
```

1.2 Users and groups

User	Groups	Purpose
admin	sudo	Administrator, allowed to use sudo
developer	developers	Modifies application files
support	support	Reads sensitive data

```
id admin
id developer
id support
```

```
admin@secure-server:~$ id admin
uid=1001(admin) gid=100(users) groups=100(users),27(sudo)
admin@secure-server:~$ id developer
uid=1002(developer) gid=100(users) groups=100(users),1001(developers)
admin@secure-server:~$ id support
uid=1003(support) gid=100(users) groups=100(users),1002(support)
```

id admin / developer / support

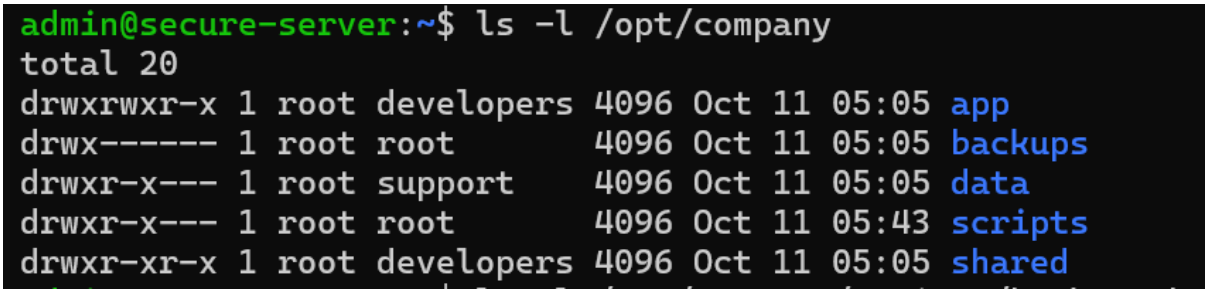
2. File Permissions and Ownership

The directory structure under /opt/company was configured according to the role of each folder.

Path	Owner:Group	Mode	Reason
/opt/company/app	root:developers	775	Developers modify files; others can only read
/opt/company/data	root:support	750	Only the support group can read sensitive data
/opt/company/backups	root:root	700	Administrators (root) only
/opt/company/scripts	root:root	750	Scripts run with high privileges; no one else may edit them
/opt/company/shared	root:developers	2770	Team folder with SGID (section 3)

2.1 Commands

```
sudo chown -R root:developers /opt/company/app
sudo chown -R root:support /opt/company/data
sudo chown -R root:root /opt/company/backups /opt/company/scripts
sudo chmod 775 /opt/company/app
sudo chmod 750 /opt/company/data /opt/company/scripts
sudo chmod 700 /opt/company/backups
```

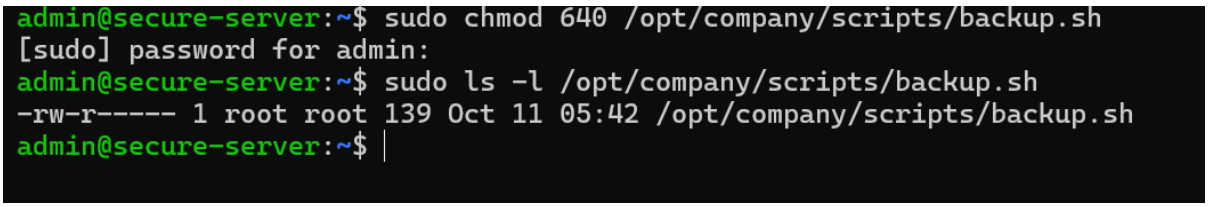


ls -l /opt/company (taken before SGID was applied to shared in section 3; shared still shows 755 here. The final 2770 value is shown in section 10).

2.2 backup.sh: numeric and symbolic chmod

```
sudo chmod 640 /opt/company/scripts/backup.sh          # numeric  -> -rw-r-----
sudo chmod u+rwx,g+rx,o-rwx /opt/company/scripts/backup.sh   # symbolic -> -rwxr-x---
```

Numeric mode sets all permission bits at once. Symbolic mode changes only the specified bits and leaves the rest untouched.



```
admin@secure-server:~$ sudo ls -ld /opt/company/scripts/backup.s
h
[sudo] password for admin:
-rwxr-x--- 1 root root 139 Oct 11 05:42 /opt/company/scripts/backup.sh
admin@secure-server:~$
```

Symbolic chmod result: -rwxr-x--- (owner rwx, group r-x, others none)

2.3 Permission tests

Test	Result
developer writes to app/	Allowed
support writes to app/	Permission denied
support reads data/	Allowed
developer reads data/	Permission denied
developer runs backup.sh	Permission denied
root runs backup.sh	Archive created in backups/

```
admin@secure-server: ~
admin@secure-server:~$ su - developer -c "touch /opt/company/app/test.txt && echo OK-developer-can-write"
Password:
OK-developer-can-write
admin@secure-server:~$ su - support -c "touch /opt/company/app/test.txt"
Password:
touch: cannot touch '/opt/company/app/test.txt': Permission denied
admin@secure-server:~$ su - support -c "ls /opt/company/data && echo OK-support-can-read"
Password:
OK-support-can-read
admin@secure-server:~$ su - developer -c "ls /opt/company/data"
Password:
ls: cannot open directory '/opt/company/data': Permission denied
admin@secure-server:~$ su - developer -c "/opt/company/scripts/backup.sh"
Password:
-bash: line 1: /opt/company/scripts/backup.sh: Permission denied
admin@secure-server:~$ sudo /opt/company/scripts/backup.sh && sudo ls -l /opt/company/backups/
Backup started: Sun Oct 11 10:31:45 UTC 2026
tar: Removing leading `/' from member names
Backup finished
total 4
-rw-r--r-- 1 root root 185 Oct 11 10:31 backup-2026-10-11.tar.gz
admin@secure-server:~$
```

3. SGID and SUID

3.1 SGID on the shared directory

```
sudo chown root:developers /opt/company/shared
sudo chmod 2770 /opt/company/shared
ls -ld /opt/company/shared        # drwxrws--- ... the 's' is the SGID bit
```

```
admin@secure-server: ~
admin@secure-server:~$ ls -ld /opt/company/shared
drwxrws--- 1 root developers 4096 Oct 11 06:26 /opt/company/shared
admin@secure-server:~$
```

Test: the same user (developer) created files in shared/ (SGID) and in app/ (no SGID).

File	Group	Explanation
shared/file1.txt	developers	Inherited from the directory because of SGID
shared/devFile.txt	developers	Inherited from the directory because of SGID
app/file2.txt	users	Primary group of the creating user

```
admin@secure-server:~$ ls -l /opt/company/shared/ /opt/company/app
/opt/company/app:
total 0
-rw-r--r-- 1 developer users 0 Oct 11 06:04 file2.txt

/opt/company/shared/:
total 0
-rw-r--r-- 1 developer developers 0 Oct 11 05:59 devFile.txt
-rw-r--r-- 1 developer developers 0 Oct 11 06:03 file1.txt
```

ls -l of shared/ and app/: files in shared/ got group developers, the file in app/ got group users.

3.2 SUID binaries

```
find / -perm -4000 -type f 2>/dev/null
```

Eleven standard system binaries were found: newgrp, mount, chsh, umount, su, chfn, passwd, gpasswd, sudo, dbus-daemon-launch-helper and ssh-keysign. No unexpected SUID file was present.

```
admin@secure-server:~$ find  / -perm -4000 -type f 2>/dev/null
/usr/bin/newgrp
/usr/bin/mount
/usr/bin/chsh
/usr/bin/umount
/usr/bin/su
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/gpasswd
/usr/bin/sudo
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
admin@secure-server:~$ ls -l /usr/bin/passwd
-rwsr-xr-x 1 root root 64152 May 30  2024 /usr/bin/passwd
admin@secure-server:~$
```

SUID list and ls -l /usr/bin/passwd (-rwsr-xr-x: the 's' is the SUID bit)

3.3 Why SUID needs special attention

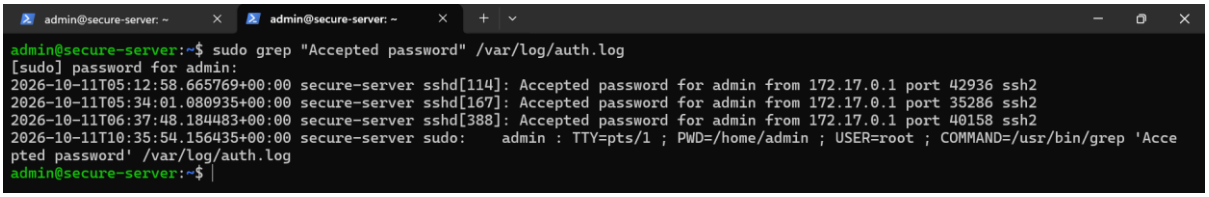
- A SUID program runs with the privileges of its owner (usually root), no matter who starts it.
- Any vulnerability in such a program (for example a buffer overflow) can give an ordinary user a root shell (privilege escalation).
- If an attacker sets SUID on a program such as bash or find, any user can become root immediately.
- Every extra SUID binary enlarges the attack surface, so the list should be minimal and reviewed regularly against a known baseline.

4. SSH Access

4.1 Password authentication

Before any hardening, SSH was first used with password authentication.

```
ssh admin@localhost -p 2222
```

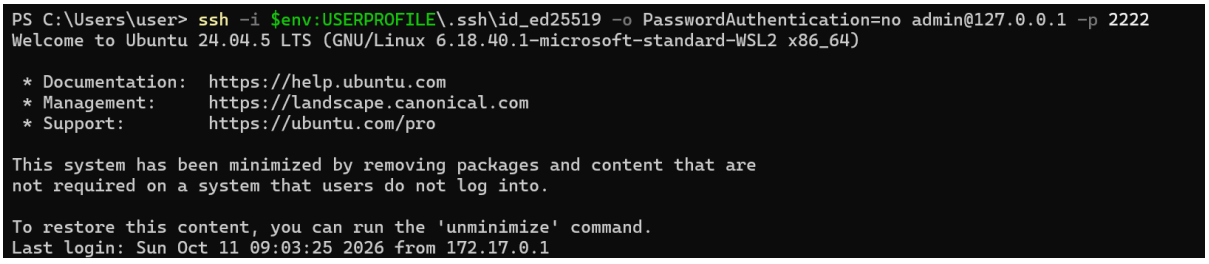


4.2 Public-key authentication

The key pair was generated on the host (client), never on the server. Only the public key was installed for admin.

```
ssh-keygen -t ed25519
type $env:USERPROFILE\.ssh\id_ed25519.pub | ssh admin@localhost -p 2222 "mkdir -p ~/.ssh
&& chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 600
~/.ssh/authorized_keys"
ssh -i $env:USERPROFILE\.ssh\id_ed25519 -o PasswordAuthentication=no admin@localhost -p
2222
```

Forcing PasswordAuthentication=no and still logging in proves that the key itself is accepted.



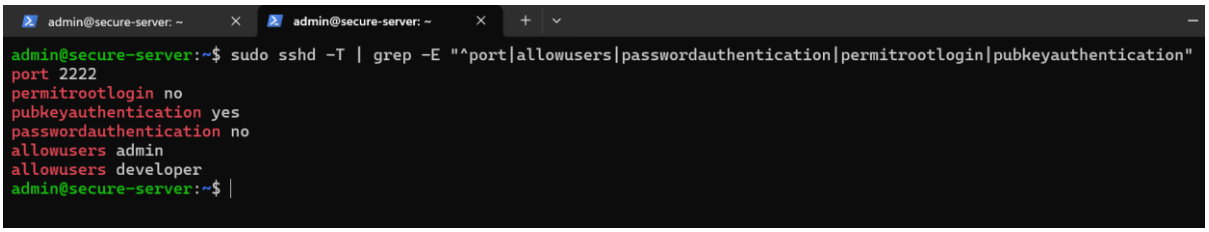
Key-only login: the command includes -o PasswordAuthentication=no and no password was requested.

5. SSH Hardening

Hardening was applied only after key authentication was confirmed working, to avoid locking the administrator out.

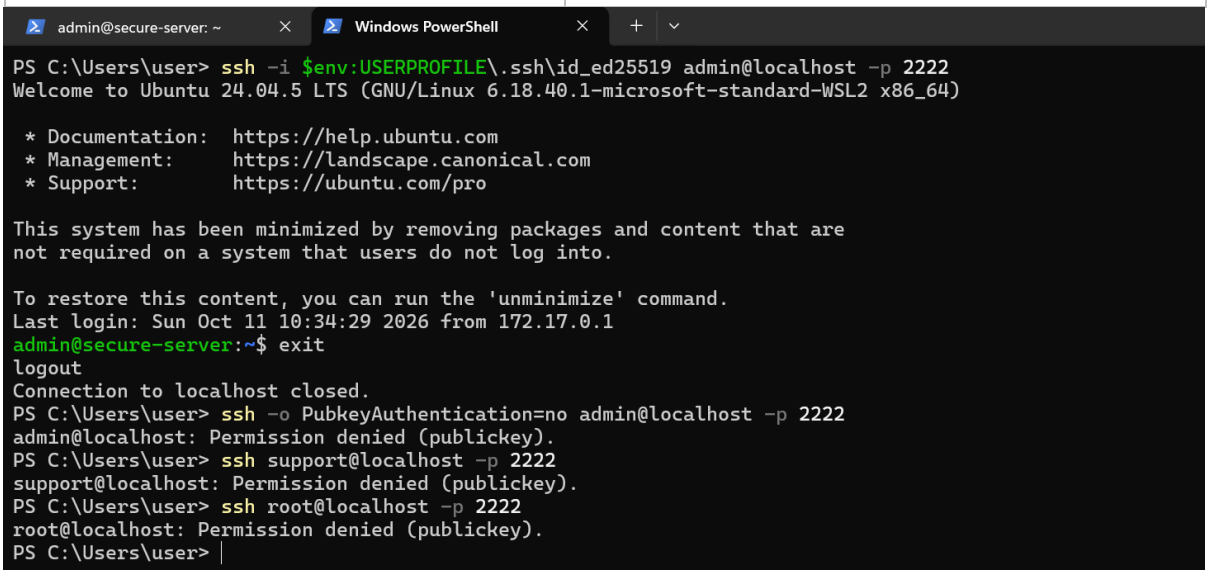
Setting	Purpose
PermitRootLogin no	Root cannot log in over SSH
PasswordAuthentication no	Passwords are rejected; keys only
PubkeyAuthentication yes	Key authentication enabled
AllowUsers admin developer	Only these users may use SSH
Port 2222	Non-default SSH port

```
sudo nano /etc/ssh/sshd_config
sudo sshd -t                        # verify syntax before restarting
sudo systemctl restart ssh
sudo sshd -T | grep -Ei
'port|permitrootlogin|passwordauthentication|pubkeyauthentication|allowusers'
```



5.1 Verification of the restrictions

Attempt	Result
admin with key	Login succeeds
admin with password only	Permission denied (publickey)
support (not in AllowUsers)	Permission denied (publickey)
root	Permission denied (publickey)



5.2 Port change on Ubuntu 24.04

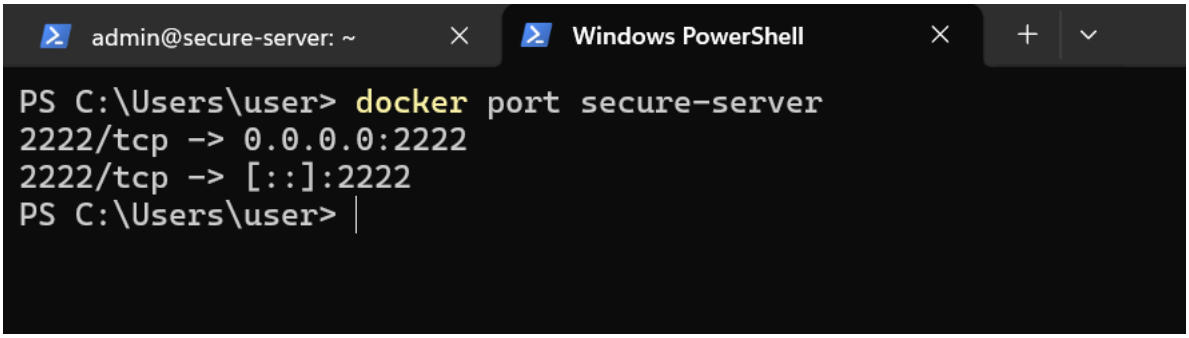
Ubuntu 24.04 starts SSH through socket activation (ssh.socket), so changing Port alone is not enough. The socket was disabled and ssh.service was enabled directly:

```
sudo systemctl disable --now ssh.socket
sudo systemctl enable ssh.service
sudo systemctl restart ssh.service
sudo ss -tlnp | grep sshd
```

```
admin@secure-server:~$ sudo ss -tlnp | grep sshd
[sudo] password for admin:
LISTEN 0      128          0.0.0.0:2222      0.0.0.0:*    users:(("sshd",pid=76,fd=3))
LISTEN 0      128             [::]:2222         [::]:*    users:(("sshd",pid=76,fd=4))
admin@secure-server:~$
```

sshd listening on 0.0.0.0:2222 and [::]:2222

The Docker port mapping was changed to -p 2222:2222 so that the host port matches the port sshd listens on inside the container.



6. Authorized Keys

The file ~/.ssh/authorized_keys lists the public keys allowed to authenticate as that user. SSH ignores it if the permissions are too open.

```
ls -ld ~/.ssh                    # drwx------ admin (700)
ls -l ~/.ssh/authorized_keys     # -rw------- admin (600)
wc -l ~/.ssh/authorized_keys     # 1 key
```

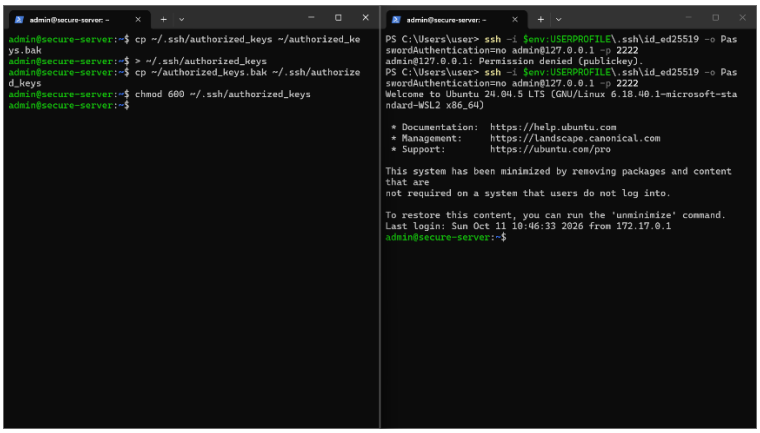
```
admin@secure-server:~$ ls -ld ~/.ssh
drwx------ 2 admin users 4096 Oct 11 06:37 /home/admin/.ssh
admin@secure-server:~$ ls -l ~/.ssh/authorized_keys
-rw------- 1 admin users 103 Oct 11 06:37 /home/admin/.ssh/authorized_keys
admin@secure-server:~$ wc -l ~/.ssh/authorized_keys
1 /home/admin/.ssh/authorized_keys
admin@secure-server:~$
```

Permissions and ownership of .ssh and authorized_keys, and one key in the file

6.1 Behaviour tests

Scenario	Observed result
Valid key	Login succeeds without a password; Accepted publickey is logged
Key removed from authorized_keys	Permission denied (publickey)
Incorrect permissions (chmod 777 ~/.ssh)	Permission denied (publickey); log shows: Authentication refused: bad ownership or modes for directory /home/admin/.ssh
Unauthorized user	Rejected: not listed in AllowUsers

Valid key: see the key-only login screenshot in section 4.2.




```
PS C:\Users\user> ssh -i $env:USERPROFILE\.ssh\id_ed25519 -o PasswordAuthentication=no admin@127.0.0.1 -p 2222
admin@127.0.0.1: Permission denied (publickey).
```

Permission denied (publickey) [confirm this is the test you ran; if it is the key-removed test, keep it here]

```
admin@secure-server:~$ sudo journalctl -u ssh --no-pager | tail -5
[sudo] password for admin:
Oct 11 09:18:41 secure-server sshd[275]: Accepted publickey for admin from 172.17.0.1 port 51558 ssh2: ED25519 SHA256:+KL2BYt+DEA2aJDfl9mE5eDe5QK/ZoBv/68bEiZUBFk
Oct 11 09:18:41 secure-server sshd[275]: pam_unix(sshd:session): session opened for user admin(uid=1001) by admin(uid=0)
Oct 11 09:18:45 secure-server sshd[275]: pam_unix(sshd:session): session closed for user admin
Oct 11 09:19:08 secure-server sshd[296]: Authentication refused: bad ownership or modes for directory /home/admin/.ssh
Oct 11 09:19:08 secure-server sshd[296]: Connection closed by authenticating user admin 172.17.0.1 port 46484 [preauth]
```

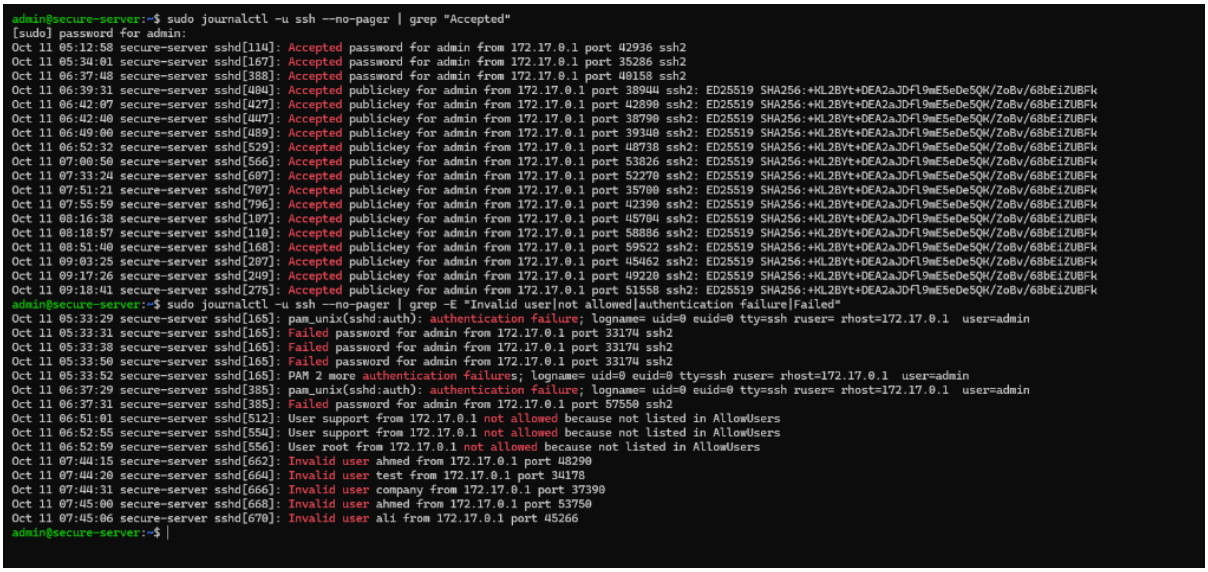
Incorrect permissions test: the log shows Authentication refused: bad ownership or modes for directory /home/admin/.ssh

Explanation: the Permission denied message that the client sees is the same in both failures. Only the server log shows the real reason.

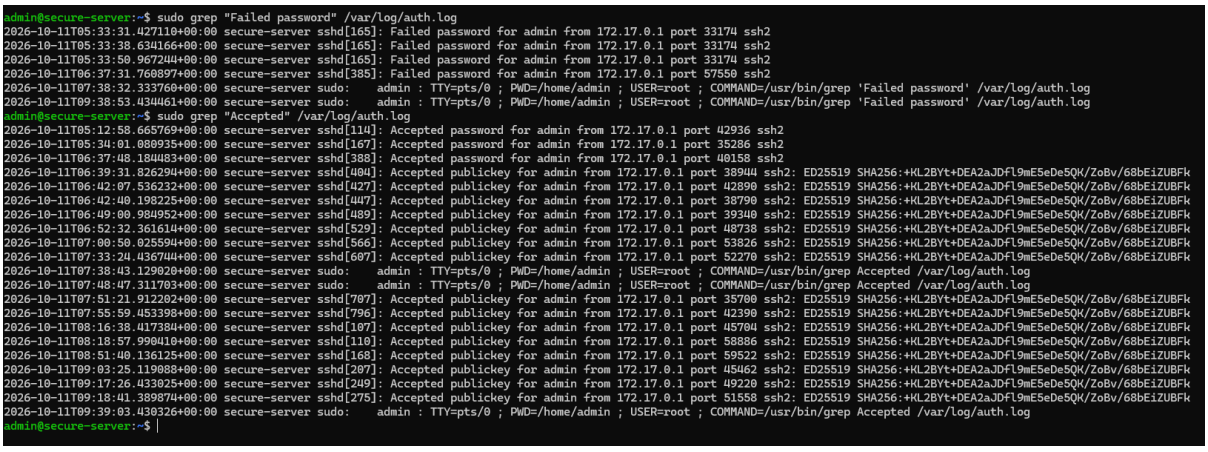
7. Log Auditing

Authentication events were investigated with journalctl and /var/log/auth.log (rsyslog is running).

```
sudo journalctl -u ssh --no-pager | grep "Accepted"
sudo journalctl -u ssh --no-pager | grep -E "Invalid user|not allowed|authentication
failure|Failed"
sudo grep "Failed password" /var/log/auth.log
sudo grep "Accepted" /var/log/auth.log
```



journalctl: Accepted events and failed / rejected events



auth.log: Failed password and Accepted events

7.1 Findings

Event	Observation
Accepted password	admin, 3 times, before the hardening (05:12 to 06:37 UTC)
Accepted publickey	admin, 15 times from 06:39 UTC onward (journalctl), always with the same key fingerprint
Failed password	admin, 4 times (05:33 and 06:37 UTC), before password login was disabled
not allowed (AllowUsers)	support (2 attempts) and root (1 attempt), 06:51 to 06:52 UTC
Invalid user	5 attempts: ahmed (x2), test, company, ali (07:44 to 07:45 UTC)

Note: the counts in this section match the log screenshots taken at that time. Further enumeration rounds were run later (see section 8.1).

Who tried to log in, when, and did it succeed? Every attempt came from 172.17.0.1, the Docker bridge gateway (the host machine). The only successful logins were the legitimate admin account.

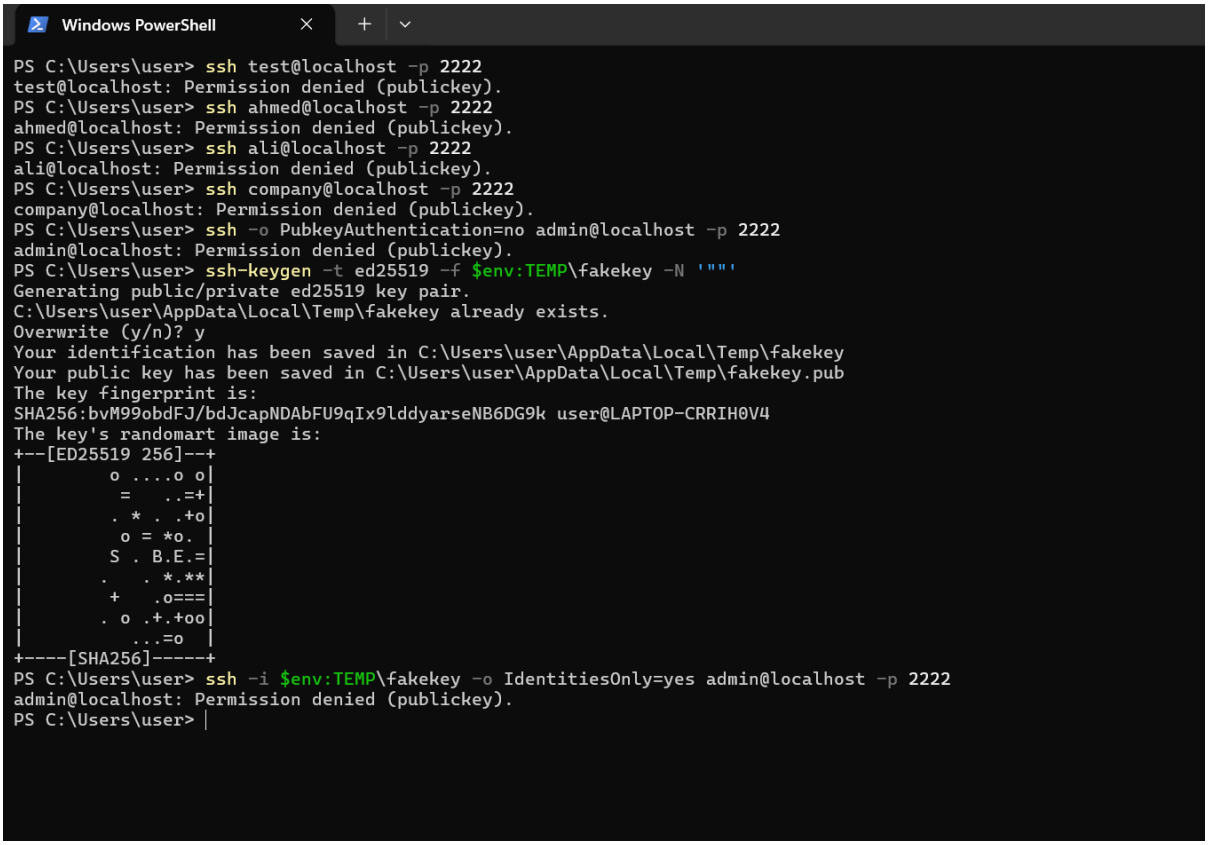
Note: in a real server the source would be the attacker's real IP address, which is what is used for blocking (firewall or fail2ban). Every sudo command is also logged in auth.log, which supports accountability.

Note: the sudo lines in the auth.log output are the audit commands themselves being logged, not authentication failures.

8. Brute-Force Attack Simulation

Failed attempts were generated from the host: wrong usernames, a wrong password and a wrong key.

```
ssh test@localhost -p 2222
ssh ahmed@localhost -p 2222
ssh ali@localhost -p 2222
ssh company@localhost -p 2222
ssh -o PubkeyAuthentication=no admin@localhost -p 2222          # wrong password attempt
ssh-keygen -t ed25519 -f $env:TEMP\fakekey -N '""'
ssh -i $env:TEMP\fakekey -o IdentitiesOnly=yes admin@localhost -p 2222   # wrong key
```



Log analysis: count of events per username.

```
admin@secure-server:~$ sudo grep "sshd.*Invalid user" /var/log/auth.log | awk '{print $6}' | sort | uniq -c | sort -rn
      4 test
      3 ahmed
      2 company
      2 ali
```

Count per username. Note: support and root come from the earlier hardening tests (section 5.1), the other four from this simulation.

8.1 Incident report

Attack detected	Yes: username enumeration and attempts to use unauthorized accounts
Source	172.17.0.1 (Docker bridge gateway, i.e. the trainee's host)
Username targeted	ahmed, test, company, ali (non-existent); support and root (blocked by AllowUsers); admin (wrong password/key)
Number of failures	Invalid user: 11 (test x4, ahmed x3, company x2, ali x2); not allowed: 3 (support x2, root x1); admin wrong password/key: 2 (Connection closed by authenticating user admin [preauth]); Failed password for admin: 4 (before hardening)
Time	2026-10-11 UTC: 05:33 to 06:52 (earlier tests); enumeration rounds at 07:44 to 07:45, 09:49 to 09:50, and 10:49 to 10:50
Successful login detected?	No successful login for any attacked account. The only successes were the legitimate admin logins (password, then key).

Note: after PasswordAuthentication was set to no, a wrong password is rejected as Permission denied (publickey) and no longer produces a Failed password event. The Failed password events in the log date from before the hardening.

9. Service Minimization

```
systemctl list-units --type=service --state=running --no-pager
ps aux
sudo ss -tulpn
```

```
admin@secure-server:~$ systemctl list-units --type=service --state=running --no-pager
  UNIT                    LOAD   ACTIVE SUB     DESCRIPTION
  dbus.service            loaded active running D-Bus System Messag…
  getty@tty1.service      loaded active running Getty on tty1
  rsyslog.service         loaded active running System Logging Serv…
  ssh.service             loaded active running OpenBSD Secure Shel…
  systemd-journald.s…     loaded active running Journal Service
  systemd-logind.ser…     loaded active running User Login Manageme…
  systemd-resolved.s…     loaded active running Network Name Resolu…
  user@1001.service       loaded active running User Manager for UI…

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

8 loaded units listed.
admin@secure-server:~$
```

Running services (8 units)

```
admin@secure-server:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  21188 12708 ?        Ss   05:07   0:01 /sbin/init
root          24  0.0  0.5  66700 20368 ?        S<s  05:07   0:01 /usr/lib/systemd/systemd-journald
systemd+      69  0.0  0.3  21348 13132 ?        Ss   05:07   0:00 /usr/lib/systemd/systemd-resolved
message+      78  0.0  0.1   9568  5416 ?        Ss   05:07   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
root          83  0.0  0.2  18184  8652 ?        Ss   05:07   0:00 /usr/lib/systemd/systemd-logind
root          86  0.0  0.0   3128  2052 tty1     Ss+  05:07   0:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog        94  0.0  0.1 222524  5712 ?        Ssl  05:07   0:00 /usr/sbin/rsyslogd -n -iNONE
root         528  0.0  0.2  12032  8188 ?        Ss   06:52   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root         707  0.0  0.2  14744 10524 ?        Ss   07:51   0:00 sshd: admin [priv]
admin        711  0.0  0.2  19816 11032 ?        Ss   07:51   0:00 /usr/lib/systemd/systemd --user
admin        712  0.0  0.0  21184  3640 ?        S    07:51   0:00 (sd-pam)
admin        728  0.0  0.1  14968  7208 ?        S    07:51   0:00 sshd: admin@pts/0
admin        729  0.0  0.1   5028  4364 pts/0    Ss   07:51   0:00 -bash
admin        737 50.0  0.1   8296  4320 pts/0    R+   07:52   0:00 ps aux
```

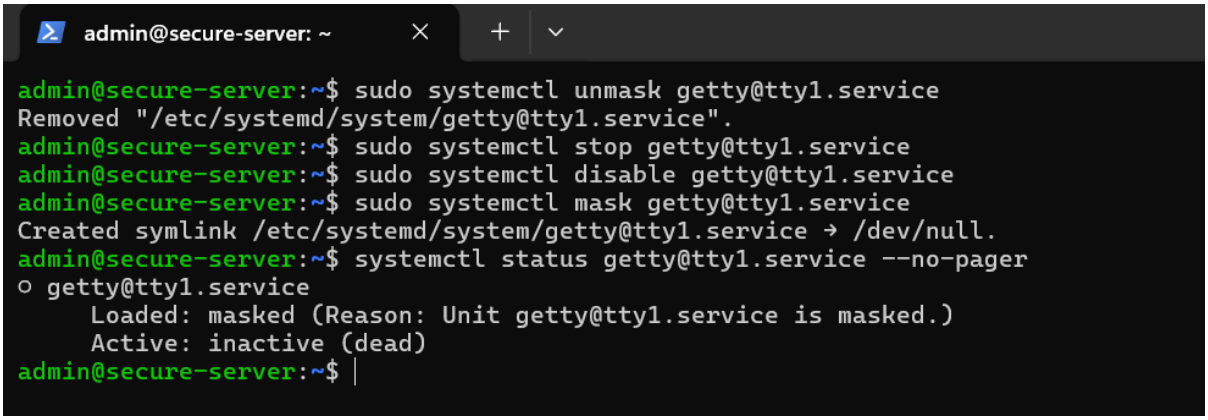
ps aux

```
admin@secure-server:~$ sudo ss -tulpn
[sudo] password for admin:
Sorry, try again.
[sudo] password for admin:
Netid State  Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp   UNCONN 0      0         127.0.0.54:53       0.0.0.0:*     users:(("systemd-resolve",pid=69,fd=16))
udp   UNCONN 0      0      127.0.0.53%lo:53       0.0.0.0:*     users:(("systemd-resolve",pid=69,fd=14))
tcp   LISTEN 0      4096      127.0.0.54:53       0.0.0.0:*     users:(("systemd-resolve",pid=69,fd=17))
tcp   LISTEN 0      4096   127.0.0.53%lo:53       0.0.0.0:*     users:(("systemd-resolve",pid=69,fd=15))
tcp   LISTEN 0      4096         0.0.0.0:22       0.0.0.0:*     users:(("sshd",pid=528,fd=3),("systemd",pid=1,fd=49))
tcp   LISTEN 0      4096            [::]:22          [::]:*     users:(("sshd",pid=528,fd=4),("systemd",pid=1,fd=50))
admin@secure-server:~$
```

Listening ports: only sshd is reachable from outside; systemd-resolved listens on localhost only

Service	Decision	Reason
ssh	Keep	The purpose of the server
rsyslog	Keep	Writes auth.log used for auditing
systemd-journald, systemd-logind, dbus	Keep	Core system components; SSH sessions depend on them
systemd-resolved	Keep (documented)	Listens on 127.0.0.53 / 127.0.0.54 only; not reachable from outside
getty@tty1	Disabled and masked	No physical console in a container

```
sudo systemctl stop getty@tty1.service
sudo systemctl disable getty@tty1.service
sudo systemctl mask getty@tty1.service
systemctl status getty@tty1.service --no-pager    # Loaded: masked
```



9.1 stop vs disable vs mask

Command	Effect	When to use it
stop	Stops the service now; it returns after a reboot	Temporary maintenance or a quick test
disable	Removes it from boot; it can still be started manually or by a dependency	A service not needed at boot but possibly needed later
mask	Links the unit to /dev/null; it cannot be started at all	A service that must never run

Result: the only port exposed to the network is SSH.

10. Security Baseline Audit

```
cat /etc/passwd | grep -E "admin|developer|support"
grep -E "sudo|developers|support" /etc/group
sudo sshd -T
sudo ss -tulpn
find / -perm -4000 -type f 2>/dev/null
find / -xdev -type f -perm -0002 2>/dev/null
sudo journalctl -u ssh --no-pager | tail -20
ls -l /opt/company/ ; ls -ld /opt/company/shared
```

```
admin@secure-server:~$ cat /etc/passwd | grep -E "admin|developer|support"
admin:x:1001:100::/home/admin:/bin/bash
developer:x:1002:100::/home/developer:/bin/bash
support:x:1003:100::/home/support:/bin/bash
admin@secure-server:~$ grep -E "sudo|developers|support" /etc/group
sudo:x:27:ubuntu,admin
developers:x:1001:developer
support:x:1002:support
admin@secure-server:~$
```

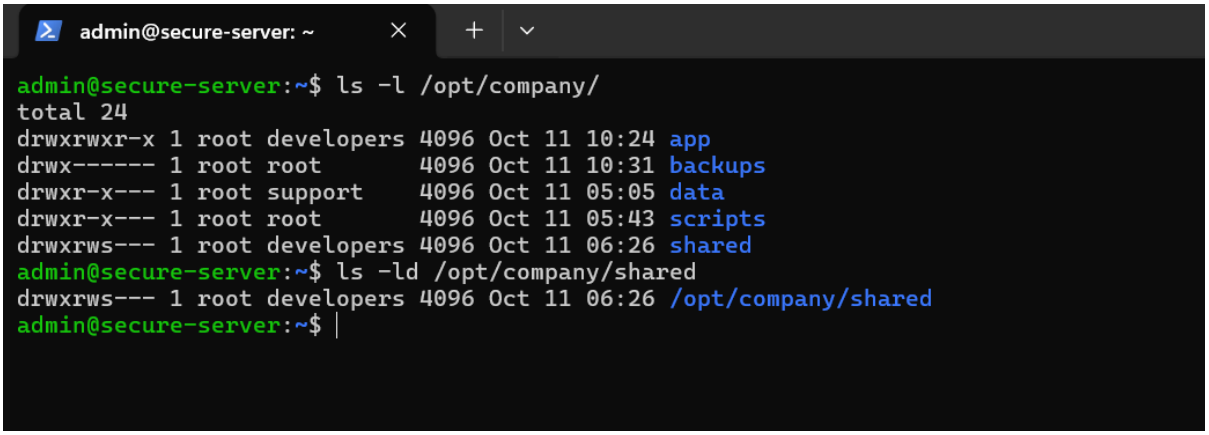
Users and groups

```
admin@secure-server: ~
admin@secure-server:~$ sudo sshd -T | grep -E "^port|allowusers|passwordauthentication|permitrootlogin|pubkeyauthentication"
port 2222
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
allowusers admin
allowusers developer
admin@secure-server:~$ sudo ss -tulpn
Netid  State   Recv-Q  Send-Q    Local Address:Port     Peer Address:Port  Process
udp    UNCONN  0       0            127.0.0.54:53           0.0.0.0:*      users:(("systemd-resolve",pid=63,fd=16))
udp    UNCONN  0       0         127.0.0.53%lo:53           0.0.0.0:*      users:(("systemd-resolve",pid=63,fd=14))
tcp    LISTEN  0       4096      127.0.0.53%lo:53           0.0.0.0:*      users:(("systemd-resolve",pid=63,fd=15))
tcp    LISTEN  0       128             0.0.0.0:2222         0.0.0.0:*      users:(("sshd",pid=76,fd=3))
tcp    LISTEN  0       4096         127.0.0.54:53           0.0.0.0:*      users:(("systemd-resolve",pid=63,fd=17))
tcp    LISTEN  0       128                [::]:2222            [::]:*      users:(("sshd",pid=76,fd=4))
admin@secure-server:~$
```

```
admin@secure-server:~$ find / -perm -4000 -type f 2>/dev/null
/usr/bin/newgrp
/usr/bin/mount
/usr/bin/chsh
/usr/bin/umount
/usr/bin/su
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/gpasswd
/usr/bin/sudo
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
```

SUID files

```
admin@secure-server: ~
admin@secure-server:~$ find / -xdev -type f -perm -0002 2>/dev/null
admin@secure-server:~$ sudo journalctl -u ssh --no-pager | tail -20
Oct 11 10:43:57 secure-server sshd[673]: Connection closed by authenticating user admin 172.17.0.1 port 38100 [preauth]
Oct 11 10:44:32 secure-server sshd[677]: Accepted publickey for admin from 172.17.0.1 port 46522 ssh2: ED25519 SHA256:+KL2BYt+DEA2aJDfl9mE5eDe5QK/ZoBv/68bEi
ZUBFk
Oct 11 10:44:32 secure-server sshd[677]: pam_unix(sshd:session): session opened for user admin(uid=1001) by admin(uid=0)
Oct 11 10:46:33 secure-server sshd[696]: Accepted publickey for admin from 172.17.0.1 port 42012 ssh2: ED25519 SHA256:+KL2BYt+DEA2aJDfl9mE5eDe5QK/ZoBv/68bEi
ZUBFk
Oct 11 10:46:33 secure-server sshd[696]: pam_unix(sshd:session): session opened for user admin(uid=1001) by admin(uid=0)
Oct 11 10:47:46 secure-server sshd[720]: Connection closed by authenticating user admin 172.17.0.1 port 53960 [preauth]
Oct 11 10:48:02 secure-server sshd[724]: Accepted publickey for admin from 172.17.0.1 port 49512 ssh2: ED25519 SHA256:+KL2BYt+DEA2aJDfl9mE5eDe5QK/ZoBv/68bEi
ZUBFk
Oct 11 10:48:02 secure-server sshd[724]: pam_unix(sshd:session): session opened for user admin(uid=1001) by admin(uid=0)
Oct 11 10:49:25 secure-server sshd[747]: Invalid user test from 172.17.0.1 port 46260
Oct 11 10:49:25 secure-server sshd[747]: Connection closed by invalid user test 172.17.0.1 port 46260 [preauth]
Oct 11 10:49:31 secure-server sshd[749]: Invalid user ahmed from 172.17.0.1 port 58104
Oct 11 10:49:31 secure-server sshd[749]: Connection closed by invalid user ahmed 172.17.0.1 port 58104 [preauth]
Oct 11 10:49:37 secure-server sshd[751]: Invalid user ali from 172.17.0.1 port 58116
Oct 11 10:49:37 secure-server sshd[751]: Connection closed by invalid user ali 172.17.0.1 port 58116 [preauth]
Oct 11 10:49:45 secure-server sshd[753]: Invalid user company from 172.17.0.1 port 40190
Oct 11 10:49:45 secure-server sshd[753]: Connection closed by invalid user company 172.17.0.1 port 40190 [preauth]
Oct 11 10:49:57 secure-server sshd[755]: Connection closed by authenticating user admin 172.17.0.1 port 39172 [preauth]
Oct 11 10:50:18 secure-server sshd[757]: Connection closed by authenticating user admin 172.17.0.1 port 44882 [preauth]
Oct 11 10:52:03 secure-server sshd[759]: Accepted publickey for admin from 172.17.0.1 port 46326 ssh2: ED25519 SHA256:+KL2BYt+DEA2aJDfl9mE5eDe5QK/ZoBv/68bEi
ZUBFk
Oct 11 10:52:03 secure-server sshd[759]: pam_unix(sshd:session): session opened for user admin(uid=1001) by admin(uid=0)
admin@secure-server:~$
```



Check	Result
Users and groups	admin (sudo), developer (developers), support (support)
SSH configuration	Port 2222, no root login, key authentication only, AllowUsers admin developer
Listening ports	SSH 2222 (external); DNS stub on localhost only
SUID files	11 standard system binaries, nothing unexpected
World-writable files	None found
Authentication logs	Failed and rejected attempts recorded and traceable

11. Final State and Conclusion

Objective	Status
Users and groups configured	Done
Sensitive files secured with ownership and permissions	Done
Numeric and symbolic chmod demonstrated	Done
SGID configured, SUID investigated	Done
Password and public-key SSH configured	Done
Root login and password authentication disabled	Done
SSH moved to a non-default port (2222)	Done
authorized_keys secured (700 / 600)	Done
Failed attempts generated and investigated in the logs	Done
Unnecessary services reduced	Done
Final audit documented	Done

Lessons learned

- Never disable password authentication before confirming that key authentication works.
- Always run sshd -t before restarting SSH, and keep an existing session open while testing.
- A Permission denied message does not say why; the log (for example not allowed because not listed in AllowUsers) does.
- Emergency access to the container without SSH: docker exec -it secure-server bash.
- On Docker Desktop, a stuck port mapping was fixed by quitting Docker Desktop, running wsl --shutdown and starting the same container again.